



NATIONAL SEMINAR

ON

**EMERGING AGRICULTURE TECHNOLOGY FOR
ENHANCED PRODUCTION & PRODUCTIVITY
OF CROPS**

13th October, 2023



Organised By:

**FACULTY OF AGRICULTURE
RKDF UNIVERSITY**

AIRPORT BYPASS ROAD, GANDHI NAGAR, BHOPAL M.P.



1	Chief Patron	Dr. (Smt.) Sadhna Kapoor, Chancellor RKDF University, Bhopal
2	Patron	Mr. Siddharth Kapoor, Pro- Chancellor, RKDF University & MD, RKDF Group
3	Chairperson	Prof. Vijay K. Agrawal, Vice Chancellor, RKDF University
4	Advisors	Dr. B. N. Singh, DGM RKDF University Prof. V. K. Sethi, DGR RKDF University Dr. N. K. Lariya, Registrar, RKDF University Dr. Sunil Patil, Examination Controller, RKDF University Dr. Ratnesh Jain, DSW, RKDF University
5	Convener	Dr. K. C. Pandey, Dean, Agricultural Engineering and Faculty of Agriculture, RKDF University
6	Co-Conveners	Dr. Suchi Gangwar, Associate Professor and Head, Faculty of Agriculture, RKDF University Dr. Shiv Singh Basediya, Associate Professor and Head (Agricultural Engineering)
7	Organizing Committee	Dr. M. L. Kori, Director Research & Dean, Faculty of Pharmacy Dr. Arun Patel, Dean, Faculty of Engineering & Technology Dr. Abhijeet B. Patil, Principal, Ram Krishna College of Ayurveda & Medical Sciences Dr. Anoop J. Katyan, Dean, RKCHMS, RKDF University Dr. C. B. S. Dangi, Professor & Dean, Faculty of Science & Paramedical Dr. M. S. Pawar, Dean, Faculty of Education Dr. Vandana Raghuvanshi, Principal, University College of Nursing Dr. Anshuma Upadhyay, I/C Dean, Faculty of Law Ar. Vibha Joshi, Dean (Design Chair), Faculty of Architecture Dr. Pratyush Tripathi, Dean, Faculty of Management Dr. N. K. Shrivastva, Dean, Faculty of Commerce & Social Science
8	Event Coordinator	Mr. Neeraj Jain, Faculty of Agriculture, RKDF University Mr. Vivek Gumasta, Faculty of Agriculture, RKDF University Dr. (Ms.) Meenakshi Samartha, Faculty of Agriculture, RKDF University Ms. Charu Bhagat, Faculty of Agriculture, RKDF University Mr. Peeyush Shrivastva, Faculty of Agriculture, RKDF University Ms. Kratika Alawe, Faculty of Agriculture, RKDF University Mr. Ravi Kumar, Faculty of Agriculture, RKDF University Mr. Rudra Pratap Singh Gurjar, Faculty of Agriculture, RKDF University Ms. Megha Khare, Faculty of Agriculture, RKDF University Dr. (Ms) Swarna Kurmi, Faculty of Agriculture, RKDF University Dr. Prashant Singh Kourav, Faculty of Agriculture, RKDF University Ms. Renu Chaturvedi, Faculty of Agriculture, RKDF University Md. Azhar Beig, Faculty of Agriculture, RKDF University Ms. Neelu Yadav, Faculty of Agriculture, RKDF University Mr. Ravi Singh, Faculty of Agriculture, RKDF University
9	Technical Team	Mr. Prashant Pare Mr. Manoj Agrawal Mr. Shaikh Faruque Mr. Vikram Shukla

Glimpse of RKDF University, Bhopal

RKDF University Gandhi Nagar, Bhopal was established by Ayushmati Education and Social Society, by an act of Govt. of M.P and published in State Gazette on 19th July 2011. The University is spread over 107 acres of land near the Bhopal Airport having buildup area of 30.266 acres and well maintained 14.485 acres Garden in campus. The University was approved by UGC under section 2(f) 1956 after the visit of expert committee members constituted by University Grant Commission, New Delhi.

After the approval of Statutes & ordinances of the University by the Department of Higher Education, Govt. of M.P. on 4th May 2012, the Academic Session started from the year 2012- 13. All the courses run by the University have approval of concern Regulatory bodies & State Govt. also. From Academic year 2013-14, the University had introduced compulsory courses of 4 Credit on “Value Spirituality and Consciousness Development” in collaboration with Brahma Kumari’s Education wing, Mount Abu, Rajasthan for all the students of undergraduate programmes which they have to pass to get a degree. The University aims to produce youths having ethical values for the Nation.

The University has established, first Carbon Capture plant (CCP) in Asia and developed modified Bukhari for army for which the university has been recognized at international forums. Biodiesel plant has also been established at the university. After valuation of achievements of the University at International level in the Academic and Research. Hon’ble Chancellor of the University, Dr. Sadhna Kapoor was awarded Honorary Professor of Oxford Academic Union, U.K. in 2017 at Switzerland.

The University is one of the fast growing Universities of central India offering a wide range of undergraduate, postgraduate and doctoral degree programs in Engineering, Pharmacy, Management, Education, Science, Commerce, Computer, Law, Agriculture, Agriculture Engineering, Architecture, Paramedical, Nursing, Homoeopathic, Ayurvedic and Humanities. Presently, the University is excelling in the leadership of visionary Vice Chancellor Prof. Vijay K. Agrawal.

About the Department

Faculty of Agriculture, RKDF University, Gandhi Nagar, Bhopal was established in year 2014. The department offered two programmes i.e. B. Sc. (Hons.) Agriculture and B.Tech (Agricultural Engineering) course curriculum. 20.77 Acres land is allotted to Faculty of Agriculture for Physical Infrastructure, Building, Hostels, Resident quarters and conducting activities on Crop cafeteria, ELP Modules, Polyhouse and Vermi-Composting at University main campus. Some innovative Projects are prepared by department like Plant shifting Robot, Multi-Purpose Agri-robot and Solar up-draft tower.

Department also won the awards at national and state level like Technology inoculation of dental conduction for hearing impaired person, Krishi Vikash Awards for Agricultural models SRIZAN-2019 and recently 2nd position in 10th Bhopal Vigyan Mela organised by Vigyan Bharti & MPCST for working model on Precision Farming.

About the State

Madhya Pradesh can be described as the best state of the nation when it comes to the diversity of its tourism destinations. Madhya Pradesh is known as the “*Heart of India*” because of a lot many things that it has to offer. It has been home to cultural and spiritual heritage of almost all the religions. Innumerable monuments, intricately carved temples, stupas, forts and palaces are dotted all over the State. The natural beauty of the state is equally varied. Consisting largely of a plateau, the State has everything, from mighty mountain ranges to meandering rivers to miles of verdant forests. The temples of Khajuraho are India’s unique gift to the world, representing love and joys of life in a sublime expression. Madhya Pradesh is also a shopper’s paradise with beautiful Maheshwari and Chanderi textiles and bell metal craft of Bastar. The state is home to three UNESCO World Heritage Sites namely Sanchi, Bhimbetka and Khajuraho.

Madhya Pradesh is the second largest producer of food grains in the country. The state is expected to produce 352.7 lakh tonnes of wheat in 2022-23 as compared to 174.8 lakh tonnes in 2013-14. Similarly, the production of rice has increased merely from 53.6 lakh to 131.8 lakh tonnes during this period. There has been a general increase in the irrigated area of the state in the past years. In the year 2021-22, the net irrigated area was 12881.6 thousand hectare, which is 2.93 percent more than last year’s 12515.2 thousand hectare. The state is making effort to diversify agricultural production by bringing more and more area under horticulture crops, fruits, vegetables, spices, etc. Production of major vegetables crops, fruits, spices and flowers have been 224.45, 90.19, 50.90 and 4.27 lakh metric tonnes, respectively in the year 2021-22.

The Madhya Pradesh has been ranked first in citrus, orange, gram, oilseeds, pulses, soybean and tomato production in the country. The state has second ranking in wheat, maize, green pea, onion and guava production. The production of coriander, chilli, mustard and milk brought the state on third position in the country. The state has been winning “Krishi Karmad Award” continuously from last Seven years.



About the Conference

The agriculture is the backbone of the country. Traditionally India had been an agricultural country. Numerous developments in the field of agriculture have taken place in past few years. There is urgent need to increase the awareness of the students and farmers about the innovations taken place in the country so that the motto of doubling the farmer income may be achieved.

There are five basic inputs of agriculture namely Seed, Fertilizer, Irrigation, Interculture and Machinery. Appropriate use of these inputs can make the agriculture more economical and profitable. Due to several initiatives taken by government and intensive efforts of scientists of ICAR and other research institutions, for increasing production and productivity of agricultural sector, the agricultural production has increased substantially. Innovations of several ICAR Institutions have transformed our country from importing food grains to surplus food grain production. Many technological innovations have taken place with passage of time like Neem Coated Urea, Nano Urea and Nano DAP have been introduced to reduce the wastage and increase the production. Natural and Organic farming is being practiced in the country with more vigor. However, there is urgent need to update the knowledge of masses working and intending to move towards field of agriculture. Also, there is utmost need to educate the students with recent developments taken place in field of agriculture to update their knowledge. The new techniques developed in the field of agriculture are of prime importance to feed the bulging population (140 crores) of our country. The adoption of new agro-techniques may certainly lead to fulfilling the food needs of our country.



The proposed conference is an attempt to update the knowledge of students with recent technological developments undertaken in agriculture field leading to enhanced production & productivity of various crops.

Program Schedule

S. No.	Keynote Speaker	Topic
1	Dr. Sandip Gangil , Head, Agricultural Energy and Power Division & Principal Scientist, ICAR-Central Institute of Agricultural Engineering, Bhopal.	Application of Nanotechnology for enhanced income of farmers.
2	Dr. K.V.R. Rao , Head, Irrigation and Drainage Engineering Division, ICAR-Central Institute of Agricultural Engineering, Bhopal.	Protected Cultivation practices for increased production and productivity.
3	Dr. A K Patra , Ex- Director and Emeritus Professor ICAR-Indian Institute of Soil Science, Bhopal.	Technology landscape for enhancing soil health in Indian agriculture

Scheduled Date: October 13, 2023

Venue: Seminar Hall, Ist Floor, Administrative Block, RKDF University, Bhopal

Programme Details

Inaugural Session

Chairman	Dr. Vijay K. Agrawal, Vice Chancellor, RKDF University
Key note Speaker	Dr. Sandip Gangil, Head, Agricultural Energy and Power Division & Principal Scientist, ICAR-Central Institute of Agricultural Engineering, Bhopal. Dr. K.V.R. Rao, Head, Irrigation and Drainage Engineering Division, ICAR-Central Institute of Agricultural Engineering, Bhopal. Dr. A K Patra, Ex- Director and Emeritus Professor ICAR-Indian Institute of Soil Science, Bhopal.
Co-Chairman	Dr. N. K. Lariya, Registrar, RKDF University, Bhopal
Convener	Dr. K. C. Pandey, Dean, Agricultural Engineering and Faculty of Agriculture, RKDF University, Bhopal

11: 00 h : Lighting of lamp and Sarshwati Vandana

11:05 h : Floral Welcome of dignitaries

11:10 h : Welcome address, Dean, Faculty of Agriculture, RKDF University Bhopal

11:15 h : Inaugural address by Chairman

Inaugural Tea 11:25 h to 11:35 h

Technical Session – I (11:40 h - 13:30 h)

Theme: Use of Nanotechnology and Protected Cultivation for Increased production	
Chairman	Dr. V. K. Sethi, Director Research, RKDF University, Bhopal
Co-Chairman	Dr. V. K. Pandey, Professor, Faculty of Science, RKDF University, Bhopal
Reporteur	Er. Charu Bhagat, Faculty of Agriculture, RKDF University, Bhopal
Presentation by Key Note Speakers	
Dr. Sandip Gangil , Head, Agricultural Energy and Power Division & Principal Scientist, ICAR-Central Institute of Agricultural Engineering, Bhopal.	11:40 h -12:30 h
Dr. K. V. R. Rao , Head, Irrigation and Drainage Engineering Division, ICAR-Central Institute of Agricultural Engineering, Bhopal.	12:30 h - 13:20 h
Lunch Break	13:20 h - 14:20 h

Technical Session –II (14:30 h- 15:10 h)

Theme: Soil Health Restoration	
Chairman	Dr. B. N. Singh, Director General Management, RKDF University, Bhopal
Co-Chairman	Dr. A. C. Nayak, Dean, Satya Sai College, RKDF University, Bhopal
Reporteur	Ms. Megha Khare, Faculty of Agriculture, RKDF University, Bhopal
Presentation by Lead Speaker	
Dr. A. K. Patra , Ex- Director and Emeritus Professor ICAR-Indian Institute of Soil Science, Bhopal.	14:30 h - 15:20 h
Interaction with Lead Speaker & Students (15:20 h- 15:50 h)	
Concluding Session (15:50 h – 16:00 h)	
Felicitation of speakers	Dr. Sandip Gangil Dr. KVR Rao Dr. AK Patra
Vote of thanks	Dr. Suchi Gangwar, Faculty of Agriculture, RKDF University, Bhopal

Concluding Tea 16:00 h to 16:15 h

Registration Fees

S.NO.	Category	Participation Only
1.	Students	Rs. 150/-
2.	Researchers/Faculties of RKDF University, Bhopal	Rs. 250/-
3.	Researchers/Faculties other than RKDF University, Bhopal	Rs. 500/-

Payment details

Name of the bank: Punjab National Bank

Account Name: Dean Faculty of Agriculture

Account Number: 7548002100000566

IFSC Code: PUNB0754800

Branch: Goundhermhou, Bhopal

Link for Participation: <https://forms.gle/Jt7gBxj1fnFwjBvd9>

Contact: Dr. Suchi Gangwar-8319464350 & Dr. Shiv Singh
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For more details visit RKDF University, Bhopal website: www.rkdf.ac.in

